

HAMZA ALI

GenAI Application Engineer — RAG Pipelines, LLM-Powered Products & Evaluation Harnesses

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SUMMARY

GenAI application engineer who ships **production LLM systems, not demos**. Built and shipped two end-to-end platforms: **FinSight**, an enterprise RAG + Text-to-SQL workspace with hybrid retrieval, Cohere re-ranking, RBAC and streaming responses, and an **AI Code Review Agent**, a multi-agent PR review platform deployed as a GitHub Action. Hands-on across the full application layer — chunking and embedding strategy, prompt and context engineering, LLM API integration with auth, retry and rate-limit handling, model routing for cost control, and instrumented evaluation harnesses (RAGAS, DeepEval G-Eval, 120 automated tests). Comfortable documenting architecture and cost/latency/accuracy tradeoffs for non-technical stakeholders.

TECHNICAL SKILLS

GenAI & LLM Applications: RAG Pipelines, LLM Agents & Agentic Workflows (LangChain, LangGraph, CrewAI), Model Context Protocol (MCP), Prompt & Context Engineering, Prompt Versioning, Copilot / Chat Experiences, Enterprise Search, LLM Failure-Mode Mitigation (hallucination grounding, context-window management, token-cost control)

Retrieval & Vector Stores: Chunking Strategy, Embedding Selection, Hybrid Retrieval (vector + BM25), Two-Stage Re-Ranking (Cohere Rerank), ChromaDB, FAISS, Pinecone, Metadata Filtering, Text-to-SQL, Tree-sitter AST Indexing & Call-Graph Retrieval

LLM APIs & Model Providers: Google GenAI SDK (Gemini), OpenAI-compatible APIs, Hugging Face, Cohere, Microsoft Azure (Azure ML), REST API design, Streaming Responses (NDJSON), Auth (JWT/HS256), Rate-Limit & Quota Handling, Exponential Backoff, Model Routing

Evaluation & LLMops: RAGAS, DeepEval (G-Eval LLM-as-judge), Ground-Truth Benchmark Suites, Deterministic Evaluators, Output Logging & Quality Scoring, Pytest, ROUGE-L / Cosine-Similarity Scoring, SARIF v2.1.0 Reporting

Backend & Data: Python, SQL, C++, FastAPI, Docker & Docker Compose, Pydantic, DuckDB, SQLite (WAL), Webhook Queues, ETL, Feature Engineering, EDA, Statistical Analysis

ML & Deep Learning: PyTorch, TensorFlow, Scikit-learn, Supervised & Unsupervised Learning, Reinforcement Learning, Neural Networks

Tools: Git, GitHub, GitHub Actions, Bash, Semgrep, Bandit, Ruff, Tableau, GitHub Copilot

GENAI PROJECTS

FinSight — Enterprise RAG + Text-to-SQL Platform with Role-Based Access Control [GitHub ↗](#)

- Shipped a production-grade **enterprise search and chat experience** (React 19 + TypeScript, FastAPI, 19 REST endpoints) with an LLM-based query classifier that performs **model routing** across RAG, Text-to-SQL, greeting, and automatic SQL→RAG fallback paths — cutting unnecessary retrieval calls and token spend on trivial queries.
- Designed the full **RAG pipeline**: document chunking strategy, embedding selection via Google's embedding API, ChromaDB vector retrieval with department-level metadata filtering, and **two-stage Cohere re-ranking** before Gemini generation with real-time NDJSON token streaming to the client.
- Built a **3-store retrieval architecture** — ChromaDB for semantic search, DuckDB for structured CSV/SQL queries, SQLite for metadata — so structured questions resolve deterministically instead of being hallucinated by the LLM.
- Implemented **API auth and rate-limit resilience**: JWT (HS256, 12-hour expiry) RBAC across 6 roles and 6 permission types, plus a RetryingEmbeddings wrapper with automatic retry (10 attempts, exponential backoff capped at 120s) on transient quota errors, and self-healing DuckDB/SQLite reconciliation on startup.
- Added a **pre-LLM cross-department guard** that blocks unauthorized queries before any model call is made — eliminating both a data-leakage class and the token cost of rejected requests.
- Instrumented a two-track **evaluation harness** — 34 Pytest tests (100% pass) covering RAGAS quality metrics and 6 RBAC security tests, plus a quota-free scoring pipeline (embedding cosine similarity, BM25, ROUGE-L) reaching **0.84 answer relevancy, 0.82 answer similarity, 0.76 context recall**, and confirming zero cross-department leakage across all 6 roles.
- Containerized the full stack with Docker Compose (separate backend/frontend Dockerfiles, hot-reload, host-mounted volumes for SQLite, DuckDB and ChromaDB) and documented the architecture for handoff.

AI Code Review Agent — Autonomous Multi-Agent PR Review Platform [GitHub ↗](#)

- Architected a 3-agent **CrewAI Flows 2.0** pipeline (Senior Developer, Security Engineer, Tech Lead) with distinct reasoning skills and tools per agent, running async in parallel before a Tech Lead agent synthesizes a grounded APPROVE / REQUEST CHANGES / ESCALATE verdict.
- Drove **token cost toward zero on the hot path** with a static pre-scan layer (Semgrep, Bandit, Ruff, regex heuristics) and a Pydantic-validated YAML governance engine that enforces team coding policies with no LLM call — the model is only invoked on findings that actually need reasoning.
- Built a multi-language **Tree-sitter AST indexer** (Python, TypeScript, JavaScript, Go, Java) mapping cross-file call graphs to surface downstream callers as structured retrieval context — grounding review comments in real code relationships rather than model recall.
- Designed a deterministic confidence-scoring rubric (100 minus weighted deductions per finding) mapping BLOCKING/WARNING rule violations directly to verdict outcomes, making agent output auditable and reproducible.
- Shipped a dual-track **AI evaluation system** — 5 deterministic evaluators (confidence math, AST compilation, guardrail consistency, SARIF-schema compliance) plus **DeepEval G-Eval LLM-as-judge** metrics — validated against a 12-case ground-truth benchmark reaching 100% verdict accuracy across security, governance and architecture categories.
- Built a durable SQLite WAL-backed webhook queue with crash/orphan-job recovery behind a FastAPI gateway, delivering GitHub PR comments, suggestion diffs and OASIS SARIF v2.1.0 export — validated by an 86-test suite (100% pass) and shipped as a reusable GitHub Action.

EXPERIENCE

Data Science Intern — Advanced Telecom Services (ATS) AI Lab

Aug 2024 – Oct 2024

- Performed end-to-end EDA and feature engineering on ~1 million rows of NOAA lightning strike records (2009–2018), transforming raw geospatial and weather data into an analysis-ready dataset.
- Built an interactive Tableau heat map visualizing strike density across the US, identifying Texas and the southeastern US as the two highest-risk zones.
- Presented risk-zone findings and mitigation recommendations to non-technical stakeholders, translating statistical analysis into concrete operational recommendations — within a broader **Microsoft Azure ML** training program covering ETL and cloud ML workflows.

EDUCATION

Bachelor of Computer Science — Abbottabad University of Science and Technology

2022 – Jul 2026

Abbottabad, Pakistan

CERTIFICATIONS

CrewAI Multi-Agent Systems (Apr 2026) • RAG — DeepLearning.AI (Dec 2025) • Machine Learning Specialization — Stanford / DeepLearning.AI (Oct 2024)